Classifying Breast Cancer Diagnoses Based on Fine Needle Aspirates

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Computing environment: R

Dataset Description

The chosen dataset is Breast Cancer Wisconsin (Diagnostic) from the UC Irvine Machine Learning Repository (UCI). This dataset contains 569 observations of 31 variables in total. Of these, 30 are continuous predictor variables computed from digitized images of a fine needle aspirate (FNA) of a breast mass. These measurements describe various characteristics of the cell nuclei including texture, smoothness, concavity, symmetry, and fractal dimensions. Each feature is provided in three forms: mean, standard error, and "worst", or the mean of the three largest values of a variable. The response variable is Diagnosis, a categorical variable which takes one of two possible values: M (malignant) or B (benign). This dataset was of interest because breast cancer is the most commonly diagnosed cancer in the United States, and the leading type of cancer among women. This medical data provides the opportunity to study diagnostic classification using real biomedical imaging. Additionally, the structured format of this dataset is well-suited to exploration of classification algorithms and model interpretation.

Methodology

Data Cleaning

Little data cleaning was necessary as the dataset has no missing values or data quality issues. The only notable changes were removing the ID column and converting the Diagnosis variable to a factor with levels M and B. The class distribution is mildly unbalanced (roughly 63% benign and 37% malignant), which guided us to use stratified sampling for both train/test

split and cross validation. For all classification models, M (malignant) was treated as the positive class.

Exploratory Data Analysis

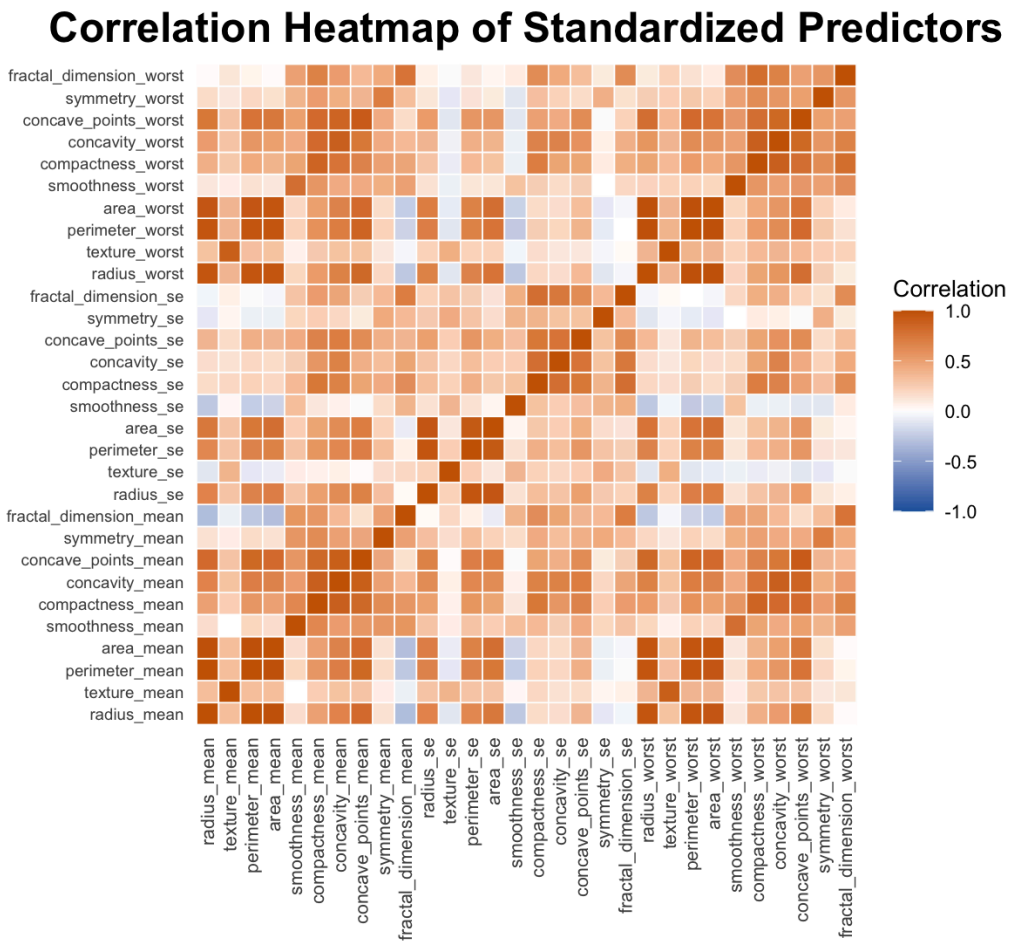

Fig. 1

Exploratory data analysis revealed multiple strong linear relationships among groups of features, particularly those measuring geometric features of cell nuclei (such as radius, perimeter, and area) across the mean, standard error, and worst measurements (Fig. 1). We did not remove or collapse these features because they are a consequence of natural biological and geometric relationships, and instead we used the presence of multicollinearity to inform further model

choices. To prepare the model for fitting, we normalized all numeric predictors by centering and scaling based on the training set, and we applied the same steps to both train and test sets to ensure a consistent basis for model performance. We then implemented an 80/20 stratified train/test split to preserve the original class proportions and trained all models stratified repeated 10-fold CV.

k-Nearest Neighbors (kNN)

We implemented a k-nearest neighbors classifier to serve as a comparison point for evaluating linear and non-linear classification boundaries. As kNN uses distance-based classification, we standardized all predictor variables using centering and scaling parameters estimated from the training set. Model tuning was conducted with a stratified repeated 10-fold cross-validation, setting the number of neighbors to maximize ROC AUC. We then evaluated the final tuned kNN model on the held-out test set to provide an additional reference point for understanding diagnostic classification.

Logistic Regression & LASSO

We fit a logistic regression model as the baseline for our analysis because it is simple and interpretable, allowing us to understand how each predictor affects a tumor's classification as malignant or benign. The model used all 30 standardized predictors to estimate the log-odds of a tumor being classified as malignant.

Next, we regularized the logistic regression model using the Least Absolute Shrinkage and Selection Operator (LASSO) technique. LASSO applies an L1 penalty to the coefficient estimates, which shrinks them toward zero and sets some exactly to zero when a predictor contributes little unique information to the model. This results in a sparser, simpler model than

standard logistic regression. The tuning parameter λ, which controls the strength of the L1 penalty applied, was selected using 10-fold cross-validation.

Random Forest

We used the caret package to train our Random Forest classifier. Since the primary evaluation metric for this project is the model's ability to classify malignant and benign cases, we based model tuning and selection on the ROC AUC measure. The tuning parameter mtry, which specifies the number of predictors randomly considered at each split, was selected using 10-fold cross validation. The value of mtry that produced the highest cross-validated value for ROC AUC was chosen as the optimal value for the final model. Using the selected mtry, we fit the final Random Forest with 500 trees to maintain stability while still having an adequate ensemble size.

Boosting & SHAP

We implemented a gradient boosting classification model using XGBoost to predict the binary diagnosis outcome. The response variable was encoded numerically, and predictor variables were scaled and converted to matrices suitable for XGBoost. An internal validation set (15% of the training data) was used for hyperparameter tuning across a grid of learning rates, tree depths, gamma values, minimum child weights, and boosting rounds, optimizing for AUC. All hyperparameter tuning was performed exclusively within the training data, and the held-out test set was used only once for final assessment. The best-performing model was evaluated on the held-out test set, with performance assessed using a confusion matrix and ROC AUC. To interpret feature contributions, SHAP (SHapley Additive exPlanations) values were calculated using XGBoost's built-in functionality, and feature importance was determined by the mean absolute SHAP value for each predictor.

Evaluation

k-Nearest Neighbors (kNN)

The kNN model performed strongly on the test set and achieved an accuracy of 0.9735, sensitivity of 0.9524, specificity of 0.9859, and an AUC of 0.9935. These results show us that even without the parametric structure used in other models, distance-based classification like kNN is able to capture much of the underlying geometry that separates benign and malignant tumors.

Logistic Regression & LASSO

For most predictors, the standard logistic regression model produced estimated coefficients that were extremely large and unstable with high standard error values. This result is consistent with the strong multicollinearity observed in Fig. 1. When evaluated on the test set, the model produced a classification error of 0.93, a sensitivity of 0.0714, a specificity of 0.0704, and an accuracy of 0.071, indicating extremely poor performance. Although the model had a high AUC of approximately 0.94, meaning it consistently scored malignant cases higher than benign ones, it classified 93% of observations incorrectly at a threshold of 0.5. This mismatch of high AUC and poor classification occurs because multicollinearity produces unstable threshold decisions.

Due to the poor performance of the logistic regression model, we next applied LASSO regularization to address issues due to multicollinearity. An optimal penalty strength of $\lambda_{min} \approx 0.0149$ was chosen. Using this λ_{min} value, LASSO kept 8 predictors and shrank the 22 remaining predictors to exactly zero. Among the selected predictors, 5 were “worst” measurements, 2 were means, and 1 was a standard error, indicating that the model relied most on extreme values of tumor measurements when classifying tumors. When evaluated on the test set, the model produced a classification error of 0.053, a sensitivity of 0.8571, a specificity of 1.000, an

accuracy of 0.95, and an AUC of 0.996, showing a significant improvement from the logistic regression model.

Random Forest

ROC was used to select the optimal model for the Random Forest classifier using the largest value. The final value chosen was mtry=2, the value with an ROC of 0.9906, sensitivity of 0.9506, and a specificity of 0.9726. After cross-validation, the final Random Forest classifier reached an AUC of 0.9925. In running the model, we achieved an accuracy of 0.9469, with a sensitivity of 0.9048 and specificity of 0.9718. These results indicate that Random Forest was effective at capturing nonlinear feature interactions and was overall accurate and stable, but the model lacks the predictive performance of the boosted model.

Boosting & SHAP

After tuning across a grid of learning rates, depths, regularization values, and boosting rounds, the best model achieved a validation AUC of approximately 0.9936. The final XGBoost classifier achieved an accuracy of about 0.9558 and a classification error of .017, delivering the strongest performance among all models evaluated. The SHAP analysis gave us a clear picture of which features the model relied on most. The predictors with the largest mean absolute SHAP values were concave points_worst, radius_worst, perimeter_worst, and area_worst, all of which consistently pushed predictions toward “malignant” when their values were high. Overall, boosting gave us both strong predictive performance and detailed interpretability through SHAP, helping us understand why the model made its predictions.

Discussion & Insights

Model	Accuracy	Sensitivity	Specificity	AUC

kNN	0.9735	0.9524	0.9859	0.9935
Logistic	0.0708	0.0714	0.0704	0.942
LASSO	0.9469	0.8571	1.000	0.9960
Random Forest	0.9469	0.9048	0.9718	0.9925
XGBoost	0.9558	0.9286	0.9718	0.9936

Fig. 2

The three strongest-performing models we observed were kNN, Random Forest, and XGBoost. These models were all capable of capturing nonlinear relationships and handling the multicollinearity naturally present in geometric measurements of FNA imaging. Standard logistic regression was the worst-performing model, showing us that solely linear decision boundaries were not useful for this data without implementing regularization. Regularization was essential for improving performance with the LASSO model, which addressed multicollinearity where the logistic model failed to do so and reduced the predictors to only highly informative predictors. Random Forest and XGBoost further improved upon the LASSO model using nonlinear methods, and boosting ended up providing the most refined decision function. The SHAP analysis also showed us that the model's most influential predictors correspond to some of the most extreme geometric values of malignant tumors, which intuitively aligns with our knowledge of the dataset.

We were able to achieve the best predictive results with ensemble methods, but there are trade-offs to these improvements. The decision-making structure of boosting is complex, even when mitigated by SHAP, and can be difficult to interpret. Linear models are less accurate, but are also more transparent and understandable for clinical users, who might want to prioritize simplicity over minor gains in accuracy. Our most accurate model is not necessarily our most

deployable model in a clinical setting, since the users need the model to be accurate and understandable.

Future extensions of our work could include exploring dimensionality reduction techniques to improve model stability without compromising prediction accuracy, or evaluating how false negatives will lead to increased clinical costs. Overall, our results emphasize the importance of carefully navigating the trade-offs between accuracy, complexity, and interpretability when building tools for medical diagnoses.